Commit-Aware Voice Agent: one-page overview

What is the project? A voice appointment agent that keeps track of what the caller now wants, what the provider actually did, and what the agent may safely say.

The example. You request maintenance at 3 PM. The provider books it but its reply is lost. You then ask for 4 PM. A timeout does not prove that 3 PM failed. Repeating the request could create another booking; stopping the assistant does not undo the first booking.

How it works. The server saves each accepted request in PostgreSQL. It records a stable operation before sending. If the result is uncertain, it checks the provider before another write. It then moves or compensates the old booking, verifies the current result and authorizes a specific confirmation. The model proposes the request; it cannot directly book or declare success.

What if old speech already played? The proposed full protocol records a persistent obligation to withdraw incompatible content and deliver a supported current update. Restarting a worker must not erase that responsibility. Sending audio is different from completed playback, and a client acknowledgement does not prove human understanding.

What is the research question? When do those additional rules protect a caller beyond a competent baseline's ordinary checks? B3 is that strong baseline; P1 adds bound speech permissions; P2 adds persistent corrective responsibility. Equal observable behaviour must receive equal grades.

What is established? Local code and PostgreSQL tests exercise revisions, unknown outcomes, permits, playback records and repair discovery. The revised six-fixture set ties at 5/6 business success in all three arms. Selected tests show extra freshness checks and explicit P2 withdrawal, not general superiority.

What is proposed next? A candidate with 500 new typed/scripted inputs: 300 clean, 100 hidden commit, 60 playback and 40 other. Three arms mean 1,500 planned executions, not 1,500 independent scenarios. All are unstarted. Calibration is separate. The target is a fixed simulator mixture, not real callers.

What remains? Supervisor review of scope, assumptions, power and observation scheduling, then a new freeze before evaluation. Real speech exposure, human understanding, general superiority and novelty remain unproven. The app is a research instrument; telephony transfer and a built-in dialer are future product work.